

Behavioral Drift in AI Systems: The Governance Problem Nobody Is Solving

*A Technical Whitepaper on Post-Deployment Behavioral Change
and the Institutional Mechanisms Required to Address It*

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Abstract

AI governance frameworks rest on a foundational assumption: that a system which passes its pre-deployment evaluations will continue to behave in the way those evaluations measured. This

paper argues that assumption is structurally wrong. AI systems drift — their outputs shift systematically over time as input distributions change, as the world changes, and as deployment contexts diverge from training conditions. We distinguish four categories of behavioral drift with distinct governance implications: covariate shift, concept drift, label (prevalence) shift, and the generative AI-specific erosion of safety properties under adversarial pressure. The machine learning community has studied these phenomena for over two decades. Governance frameworks have not kept pace. Surveying the EU AI Act, the NIST AI Risk Management Framework, ISO 42001, and FDA's Predetermined Change Control Plan framework, we find that no major regulatory instrument requires developers to specify, before deployment, what drift signals they will monitor, at what thresholds they will intervene, who authorizes that intervention, or how drift-related incidents will be documented. We identify the structural reasons this gap persists — the invisibility of slow drift, the proprietary reference problem, counterfactual attribution difficulties, and the non-statistical character of generative AI drift — and propose a minimum governance architecture for high-risk deployments. We close with five specific recommendations directed at regulators, standards bodies, and deployers in high-risk sectors.

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1. The Problem in Concrete Terms

Imagine a hospital network in a mid-sized American city that, in early 2023, deploys an AI-based clinical risk stratification system. The system ingests patient demographics, vital signs, lab values, and prior diagnoses to produce a risk score that clinicians use to triage incoming patients — deciding who gets seen first, who gets admitted, who gets the pulmonology consult. Before deployment, the system was validated on a retrospective cohort drawn from that same hospital's records going back five years. It performed well. Sensitivity and specificity were strong. Calibration was reasonable. The system cleared its pre-deployment evaluation, the vendor issued its conformity documentation, and the hospital's clinical informatics team signed off. Everyone did their jobs.

By late 2024, something has changed — but nothing that shows up in any log or alert. The city's demographics have shifted. A significant influx of patients from communities with different baseline disease prevalence, different medication histories, and different patterns of prior healthcare access has changed the composition of the patient population the model encounters every day. The model was trained on historical records that didn't reflect this demographic mix. Its outputs — those risk scores — have silently become less accurate for substantial portions of the population being triaged. High-risk patients are being scored lower than they should be. The model hasn't changed. The world has. And because nobody set up a monitoring system that would catch this, and because no governance framework required one, the degradation is invisible until someone starts asking the right questions.

This is not a hypothetical. The phenomenon of performance degradation under demographic and population shift is well-documented in the clinical AI literature. Research published in

Nature Medicine and Nature Scientific Reports has demonstrated that models trained on homogeneous historical cohorts systematically underperform when deployed into more diverse real-world populations — not because the models are maliciously biased, but because the statistical relationship between input features and clinical outcomes is not uniform across subpopulations, and training data did not capture that heterogeneity. The model that was accurate when deployed becomes less accurate over time, not because anyone changed it, but because the world it's operating in has diverged from the world it learned from.

Now ask the governance question: if the model passed every evaluation it was required to pass, and nobody changed anything, how does any current governance framework account for what went wrong? Who is responsible? What documentation was required? What monitoring should have caught it? What intervention protocol should have been triggered, and who should have authorized it? The honest answer, in 2026, for almost every major regulatory framework and standards instrument, is: we don't know, because nobody specified it. This paper is an attempt to explain why, and to describe what would have to change.

2. Defining Behavioral Drift with Precision

The word "drift" circulates through AI policy discussions in a way that is more atmospheric than analytical. People use it to mean something like "the system stopped working as well as it used to," without distinguishing among the very different mechanisms that can produce that result — mechanisms that have distinct causes, distinct signals, and very different governance implications. Getting the taxonomy right is not pedantry. If you can't say what kind of drift you're dealing with, you can't design a monitoring system for it, you can't specify intervention thresholds, and you can't assign accountability for it. So we start here.

2.1 Covariate Shift

Covariate shift occurs when the distribution of the model's inputs — $P(X)$ — changes between training and deployment, while the relationship between inputs and outputs — $P(Y|X)$ — remains stable. The model has learned a function that is still correct in principle; the problem is that the inputs it's now seeing are drawn from a different distribution than the inputs it learned from. The model isn't wrong about the world it was trained on. The world it's encountering is different.

Definition: Covariate Shift

Covariate shift occurs when $P(X)$ changes between the training distribution and the deployment distribution, while $P(Y|X)$ — the conditional relationship between inputs and outputs — remains constant. The model's learned function is still correct in principle; but the domain of inputs it encounters in deployment no longer matches the domain over which it was trained and validated. Performance degrades because the model is applied outside its effective support, not because its internal logic has become wrong.

*Classic treatment: Quiñero-Candela et al., **Dataset Shift in Machine Learning** (MIT Press, 2009).*

The clinical scenario in Section 1 is a covariate shift story. The relationship between lab values, vital signs, and clinical risk — $P(Y|X)$ — has not fundamentally changed. But the distribution of patients presenting with particular combinations of those features has shifted. A fraud detection model trained on pre-pandemic consumer transaction patterns and then deployed into the post-pandemic economy faces the same problem: the model's learned associations are still valid representations of the old behavioral landscape, but the distribution of transaction patterns the model is now scoring has moved. The model is applying a map to territory that no longer matches it.

Covariate shift is arguably the most common form of drift in production AI systems, and it is the form most amenable to statistical detection — at least in principle. Monitoring frameworks can compare the distribution of incoming inputs against the training distribution using statistical divergence measures. The challenge is that doing this well requires access to the original training distribution as a reference, which is often proprietary and rarely shared with the parties who would need to conduct an independent audit.

2.2 Concept Drift

Concept drift — what the ML literature also calls "real drift" — is a different and in some ways more fundamental problem. Here, the underlying relationship between inputs and outputs — $P(Y|X)$ itself — changes over time. What counted as a high-risk patient profile in 2022 may not count as one in 2025, because treatment protocols have changed, because new drugs have altered disease trajectories, because clinical definitions have been revised. What counted as a fraudulent transaction in a pre-contactless-payment economy may look entirely different in a world where transaction patterns have structurally evolved.

Definition: Concept Drift

Concept drift (also termed "real drift") occurs when $P(Y|X)$ changes over time — the underlying relationship between the model's inputs and the target outcome has itself shifted. The model's learned function has not merely been applied to unfamiliar inputs; its core predictions have become structurally incorrect because the world-state it modeled no longer holds. Concept drift is harder to detect than covariate shift because it cannot be identified from the input distribution alone — it requires labeled outcomes, which may arrive with significant delay or not at all.

*Survey treatment: Gama, Žliobaitė, Bifet et al., "A Survey on Concept Drift Adaptation," **ACM Computing Surveys** 46(4), 2014.*

A credit scoring model deployed in 2019 and still operating in 2025 faces concept drift on multiple axes. The economic conditions that defined "financial risk" in 2019 — employment stability patterns, debt-service ratio norms, housing market dynamics — are structurally different

from those in 2025. The model's internal representation of creditworthiness has not changed. The world's creditworthiness landscape has. In criminal justice, a recidivism prediction model trained on historical outcome data encodes the relationship between observable features and reoffending as it existed in the historical period — a relationship that reflects the policing practices, social conditions, and legal landscape of that period, all of which change.

Concept drift is more insidious than covariate shift because it is harder to detect. Detecting covariate shift requires only monitoring input distributions — no labels needed. Detecting concept drift requires observing outcomes and comparing them to predictions over time. In many high-stakes domains, labeled outcomes arrive slowly (a patient's health trajectory over months), inconsistently (criminal justice recidivism depends on rearrest, which is itself a function of policing intensity), or not at all (in systems where ground truth is never formally captured). This feedback delay is not incidental; it is structural, and any governance framework that doesn't account for it is inadequate by design.

2.3 Label Shift (Prevalence Shift)

Label shift — sometimes called prevalence shift — is conceptually simpler but practically devastating for probabilistic predictors. Here, the marginal distribution of the outcome being predicted — $P(Y)$ — changes, even while $P(X|Y)$ remains roughly constant. The base rate of the thing being predicted shifts, and a model calibrated for one base rate will produce systematically wrong probability estimates at another.

Definition: Label Shift (Prevalence Shift)

Label shift (prevalence shift) occurs when $P(Y)$ — the marginal probability of the outcome — changes between training and deployment, while the conditional distribution $P(X|Y)$ remains approximately stable. A model calibrated for outcome prevalence p_0 will produce systematically miscalibrated probability estimates when deployed in a context where prevalence is $p_1 \neq p_0$, even if its discriminative performance (e.g., AUC) is unchanged. The problem is calibration, not discrimination — and calibration is what clinicians and decision-makers rely on when acting on model outputs.

Empirical demonstration: Godau et al., "Deployment of Image Analysis Algorithms under Prevalence Shifts," arXiv:2303.12540, 2023.

Consider a radiology AI trained on a dataset where a particular pathology — say, a specific pulmonary finding — had a prevalence of 5 percent. The model is calibrated for that prevalence. A score of 0.7 from that model means something specific: given the training prevalence, there is roughly this much evidence for the finding. Now deploy that model in a context where population changes, seasonal patterns, or an emerging health situation has pushed prevalence to 15 percent. The model's discriminative performance — its ability to rank cases by risk — may be entirely unchanged. But its calibration is wrong. Its probability estimates no longer reflect the actual posterior probability of the finding given the evidence. Clinicians acting on those scores

are making decisions based on systematically incorrect risk quantification, even though nobody changed the model and the model's AUC looks fine on the next internal evaluation.

Empirical work by Godau et al. (2023) across thirty medical image classification tasks demonstrates that prevalence shifts induce substantial miscalibration in deployed classifiers, and that this miscalibration propagates into suboptimal clinical decisions in ways that standard validation metrics fail to capture. The practical governance implication is stark: a model that "passes" post-deployment evaluation using AUC or accuracy metrics may be producing systematically misleading probability estimates that are invisible to those metrics. Evaluation design that does not include calibration monitoring for prevalence shift is not adequate monitoring.

2.4 Generative AI and the Non-Statistical Drift Problem

The three drift types above fit into a shared statistical framework — they are all describable as changes to some component of a joint distribution $P(X, Y)$. Generative AI systems, and large language models in particular, introduce a different kind of behavioral change that does not map neatly onto these definitions, and for which the statistical monitoring literature offers almost nothing.

LLM-based systems can drift in at least three ways that current governance frameworks are entirely unprepared for. First, safety guardrails erode under sustained adversarial pressure. As documented in a growing body of research on jailbreaking, sufficiently persistent or creative prompt sequences can induce LLMs to produce outputs that their safety training was intended to prevent. This erosion is not a one-time event — it is ongoing, and it worsens as adversarial techniques become more sophisticated and are shared across communities. A model that passed its safety evaluation at deployment may be functionally less safe six months later, not because anything in the model changed, but because the effective attack surface has expanded through collective adversarial discovery.

Second, fine-tuning and continued training — including reinforcement learning from human feedback applied post-deployment — can shift a model's tonal calibration, its tendency toward particular kinds of outputs, and its implicit value weightings in ways that are difficult to measure and nearly impossible to attribute. The model that was deployed is not quite the same model that is operating a year later, even if the version number hasn't changed. Third, emergent capabilities — behaviors that appear at scale or under specific prompting conditions — can appear or disappear across model versions in ways that are poorly understood and not reliably tested for. A capability that a red team missed pre-deployment may emerge under deployment conditions; a safety property that was present in testing may fail to generalize to the full distribution of real-world inputs.

The governance challenge with generative AI drift is not just that it is hard to measure. It is that the conceptual framework governance relies on — a model with a defined function, producing outputs in a defined domain — doesn't fully apply. An LLM is not a function from inputs to outputs in the same sense a classifier is. Its "behavior" is a distribution over an effectively unbounded output space, and the properties we care about — safety, accuracy, calibration, tone

— are not scalar quantities with clear thresholds. Governing this requires different tools than governing a radiology classifier, and almost no current framework acknowledges the distinction.

3. Why This Is a Governance Problem, Not Just an Engineering Problem

At this point a reasonable technical reader might object: the machine learning community has known about distribution shift since at least the early 2000s. Monitoring tools exist. Statistical tests for covariate shift — maximum mean discrepancy, population stability index, Kolmogorov-Smirnov tests applied to feature distributions — are implemented in production MLOps platforms. Retraining pipelines are standard practice in sophisticated ML engineering teams. Why is this a governance problem at all? Shouldn't better engineers just solve it?

The answer requires distinguishing between the detection of drift and the governance of drift. The engineer can, in principle, detect that the input distribution has shifted by some measurable amount. What the engineer cannot determine — and what requires institutional process — is whether that shift is large enough to require intervention, who authorizes the intervention, what obligations to downstream users and affected populations are triggered by a drift event, and how the incident must be documented and communicated. None of these questions have technical answers. They are governance decisions, and they require defined institutional processes, accountability structures, and documentation standards.

Consider the clinical scenario again. Suppose the hospital's data science team does have a monitoring pipeline, and it flags that the input distribution has shifted meaningfully from the training distribution. Now what? Does the system get suspended while recalibration is performed? Who makes that call — the data scientist who ran the test, the clinical informatics director, the attending physicians, hospital administration? What is the threshold for suspension versus continued operation with added clinical scrutiny? If patients were triaged suboptimally during the period of drift, is there a notification obligation? An incident report? A disclosure to the device regulator? Under current frameworks, the answers to all of these questions are either not specified, left entirely to the deployer's discretion, or obscured across liability frameworks that have not been updated to address AI-specific failure modes.

This is the governance gap. It is not that engineers don't know drift exists — many do, and sophisticated teams monitor for it. It is that the institutional scaffolding that should surround that monitoring — defined thresholds, authorization chains, documentation standards, accountability for failures to monitor, and incident reporting mechanisms — is almost entirely absent from the regulatory instruments that are supposed to provide it. The monitoring that does exist in deployed systems is largely voluntary, internally defined, and invisible to regulators, affected populations, and external accountability mechanisms. That is not governance. That is a private engineering practice that happens to overlap with a governance need.

There is also a structural incentive problem worth naming directly. Drift monitoring, if done properly and with defined intervention triggers, creates obligations — to suspend systems, to retrain, to notify users, to document adverse events. Organizations that have not been required

to monitor for drift have no obligation to discover problems they don't look for. Voluntary monitoring frameworks create a perverse dynamic: the more rigorously a team monitors, the more obligations they generate for themselves. The organization that doesn't monitor faces no such obligations. This is a classic regulatory gap problem, and it will not be solved by calling on developers to act in good faith.

4. What the Frameworks Actually Say

We are not arguing from ignorance of existing frameworks. Each of the major governance instruments has something to say about post-deployment monitoring. What they say, when examined closely, is less than they appear to promise. Let us be specific.

4.1 EU AI Act — Article 72

Article 72 of the EU AI Act, which entered into force in August 2024, requires providers of high-risk AI systems to "establish and document a post-market monitoring system" that "actively and systematically collect[s], document[s] and analyse[s] relevant data... on the performance of high-risk AI systems throughout their lifetime." This language sounds substantial. It is less so under scrutiny. Article 72 says nothing specific about what data must be collected, how performance must be measured, what methodology is adequate, what thresholds trigger alerts, or what response protocols are required when performance degrades. The Article requires a post-market monitoring plan as part of technical documentation, but as of the date of this paper, the Commission's implementing act establishing a template for that plan — due by February 2026 — had not been finalized. The requirement, in practice, is to have a plan. The plan can say almost anything and technically comply.

More critically, Article 72 contains no mention of drift typology, no requirement to specify what kinds of distributional change the monitoring system is designed to detect, no requirement to define intervention thresholds, and no requirement to specify who is authorized to make intervention decisions. Developers subject to the EU AI Act must say that they are monitoring; they are not required to demonstrate that what they are doing constitutes meaningful monitoring by any defined standard. Article 73 covers reporting of serious incidents, but "serious incident" is defined in terms of harm that has already occurred — it does not create a pre-harm reporting mechanism for performance degradation that has not yet produced identifiable adverse events.

4.2 NIST AI Risk Management Framework

The NIST AI RMF, published in January 2023, organizes AI risk management around four functions: GOVERN, MAP, MEASURE, and MANAGE. Both GOVERN and MEASURE include language about ongoing monitoring. MEASURE 2.5 states that "AI system to be deployed... are demonstrated to be valid and reliable through the identified period of deployment" and that "AI system performance or outcomes are periodically or continuously evaluated against [organizational] defined metrics." MANAGE 4.1 addresses responding to residual risks. The framework genuinely attempts to embed monitoring into the AI lifecycle rather than treating it as an afterthought.

The fundamental limitation of the NIST AI RMF for drift governance is threefold. First, it is voluntary. Organizations can adopt it, adapt it, or ignore it. Second, its monitoring provisions describe desirable properties without specifying what constitutes adequate monitoring for any class of system — there are no sector-specific thresholds, no methodological requirements, and no accountability mechanism for organizations that claim to follow the framework but implement monitoring that is insufficient. Third, the RMF does not distinguish among drift types. Its monitoring language is generic — "performance outcomes" are to be "evaluated" — without acknowledging that covariate shift, concept drift, and prevalence shift require different detection methods and different intervention protocols. A framework that treats all performance change as equivalent cannot address the specific governance requirements of each drift type.

4.3 ISO/IEC 42001:2023

ISO/IEC 42001, published in December 2023 as the first international AI management system standard, includes requirements in Section 9 ("Performance Evaluation") for "monitoring, measurement, analysis and evaluation" of the AI management system. It requires organizations to "determine what needs to be monitored and measured" and to "evaluate the performance and the effectiveness of the AIMS." These requirements apply at the management system level — they are about whether the organization's processes for managing AI are working, not about whether specific deployed models are behaving as expected.

ISO 42001 does not specify drift monitoring at the model level. It does not require organizations to define what distributional change in model inputs or outputs would constitute a trigger for review. It does not specify calibration monitoring, incident classification, or accountability chains for model-level performance events. It is an organizational governance standard, and as such it is a reasonable foundation — but it is not a substitute for model-level behavioral monitoring requirements, and it should not be presented as one.

4.4 FDA Predetermined Change Control Plan

The FDA's Predetermined Change Control Plan (PCCP) framework, for which guidance was issued for AI-enabled device software functions in 2023 and updated in draft form in August 2024, is the most substantively developed regulatory approach to performance change in deployed AI systems. A PCCP allows developers to pre-specify, in their marketing submission, what types of changes will be made to the device — including modifications to AI/ML components — and to receive pre-authorization for those changes, avoiding the need for new 510(k) clearance with each modification. The PCCP must describe the "Description of Modifications," the "Methodology" for developing and validating changes, and an "Impact Assessment."

This is a meaningful advance, and it represents genuinely more sophisticated thinking about the AI lifecycle than any other major regulatory instrument. The PCCP framework's critical limitation, for purposes of drift governance, is that it governs intentional change, not unintended behavioral change. A PCCP pre-authorizes specific modifications that the developer plans to make. It does not require developers to specify what unintended performance degradation will trigger intervention, who authorizes that intervention, or how performance degradation that falls short of

a "serious adverse event" but above background noise should be classified, documented, and reported. Drift, by definition, is not a planned modification. The PCCP framework addresses the planned-change governance problem well; it does not address the unintended-behavioral-change governance problem at all.

4.5 Summary Comparison

The table below summarizes how each major framework addresses the core components of drift governance. The assessment reflects what each framework explicitly requires, not what organizations might choose to implement voluntarily.

Framework	Drift or Distribution Shift Explicitly Mentioned?	Monitoring Methodology Specified?	Intervention Thresholds Required?	Accountability Chain Required?	Drift Incident Reporting Required?
EU AI Act (Art. 72)	No — "performance monitoring" only	No — methodology left to developer	No — no threshold specified	No — not specified	Partial — serious incidents only (Art. 73); no pre-harm reporting
NIST AI RMF 1.0	No — generic "performance outcomes"	No — voluntary, no methodology prescribed	No — no thresholds	Partial — GOVERN function covers accountability broadly	No — voluntary framework with no reporting mandate
ISO/IEC 42001:2023	No — management system level only	No — organization defines what to measure	No	Partial — organizational governance structures required	No
FDA PCCP (AI/ML SaMD)	No — addresses planned	Partial — validation methodology	Partial — for planned	Partial — for planned	Partial — adverse event reporting

	changes, not drift	for planned changes	modifications under PCCP	modifications only	under existing MDR rules; no drift-specific reporting
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The Core Finding

No major governance framework currently requires developers to specify, before deployment: (1) which drift types are relevant to their system; (2) what monitoring signals they will track; (3) at what thresholds those signals will trigger alerts; (4) who is authorized to make intervention decisions; or (5) how drift-related performance events will be documented and communicated. What exists is the appearance of monitoring requirements — language that references "performance monitoring" without specifying what monitoring means, what it must detect, or what it must trigger. This is not governance of drift. It is governance of the idea of monitoring.

5. Why Drift Is Structurally Hard to Govern

Acknowledging the gap is easier than closing it. There are real structural reasons why drift has proven difficult to govern, and any serious proposal for better governance must engage with them honestly rather than dismissing them as excuses for inaction.

The invisibility problem. Drift, unlike a system crash or a clearly erroneous output, often manifests slowly and silently. Users don't experience a sudden failure; they experience gradually degrading outputs in ways that are hard to distinguish from the normal variation in model performance. A risk score that is systematically 8 points too low for a particular subpopulation does not announce itself. It looks like noise until someone runs the analysis to show it isn't. By the time a pattern becomes identifiable through routine use, significant harm may have accumulated — and because the degradation was gradual, the causal chain between drift and harm is hard to establish in retrospect. This makes drift governance politically uncomfortable: the harm that accumulates silently is diffuse and attributable only in aggregate, which makes it harder to convert into regulatory action than a high-profile single-incident failure.

The reference problem. Detecting drift requires a reference distribution — some representation of what the system's input space looked like at training or at deployment. This reference is almost always proprietary. The training data that defines the reference distribution is the developer's core intellectual property. Regulators and auditors who would need access to evaluate whether meaningful drift has occurred typically cannot obtain it. External researchers — the ones who might independently verify that drift is happening in a deployed clinical system — are working without access to the reference that would make detection possible. Roschewitz et al. (2024) demonstrate that effective shift identification frameworks require access to a reference dataset representative of the validation distribution; without it, distinguishing genuine distributional shift from model-specific variance is guesswork.

The counterfactual problem. When a drifted system causes harm, establishing that the drift — rather than some other factor — was responsible requires answering the counterfactual: what would the system have output on these inputs if drift had not occurred? This question is nearly impossible to answer without the original model in the original deployment context. Systems are typically retrained or updated by the time a harm event is investigated, eliminating the counterfactual comparison. This makes liability attribution genuinely difficult and gives developers a structural advantage in any post-hoc accountability proceeding: the model that caused harm is gone, and the question of what it would have done cannot be answered.

The attribution problem. Even if drift is detected and its effects are characterized, the question of who is responsible is not straightforward. The developer trained the model on historical data that is now stale — was that negligence, or the unavoidable consequence of training on any dataset? The deployer did not implement meaningful monitoring — but was monitoring required, and if so, what kind? The regulator did not specify monitoring requirements — but did the regulator have the technical capacity to do so, and who should have closed that gap? Current liability frameworks in AI — which in most jurisdictions default to product liability doctrine or general negligence principles — have no clean answer to distributed causation of this kind. Multi-party harm with no single proximate cause is precisely the scenario that existing tort frameworks handle worst.

The generative AI complication. For large language models and other generative systems, "drift" in the statistical sense is not even the right frame. The behavioral properties we care about — safety, factual accuracy, calibration, alignment with stated values — are not scalar quantities with well-defined reference distributions. Safety guardrails degrade under adversarial pressure in ways that look less like statistical distribution shift and more like a dynamic system being pushed toward an attractor state by sustained external force. Research on jailbreaking has demonstrated that safety properties that appear robust in red-team evaluations can be undermined by prompt sequences that exploit the model's training distribution in ways testers didn't anticipate. This is not covariate shift. It is not concept drift. It is something for which the ML monitoring literature offers no established framework, and for which current governance instruments have no response protocol at all.

6. What Adequate Drift Governance Looks Like

We are not proposing a utopian governance architecture that requires infinite monitoring resources and perfect regulatory capacity. We are proposing a minimum viable governance architecture for high-risk AI deployments — the floor below which a system operating in healthcare, criminal justice, financial services, or critical infrastructure should not be permitted to fall. The components of this architecture are not novel in isolation; analogues exist in pharmaceutical post-market surveillance, aviation safety incident reporting, and financial audit standards. What is novel is their combination into a coherent framework specifically designed for the behavioral drift problem in AI systems.

Pre-deployment drift specification. Before a high-risk AI system is authorized for deployment, the developer must specify what types of drift are relevant to the system and how each will be

monitored. This specification should include: (a) the drift types the system is susceptible to (covariate shift, concept drift, prevalence shift, or behavioral erosion for generative systems); (b) the specific signals that will be monitored for each type — input feature distributions, output calibration metrics, performance on demographically defined subgroups, or behavioral benchmarks for generative systems; (c) the thresholds at which monitoring signals will trigger alerts; and (d) the response protocol for each alert level. This specification becomes part of the system's conformity documentation and is available to authorized auditors. It is not a guarantee that drift will be caught — it is a commitment to a defined process that can be evaluated and audited.

Reference baseline preservation. At deployment, a standardized snapshot of the system's input distribution and output calibration must be preserved in a format accessible to authorized auditors. This does not mean making training data public — it means preserving a statistical summary of the training distribution (feature statistics, calibration curves, subgroup performance metrics) and the deployment distribution at go-live, in a standardized format that enables authorized third parties to evaluate subsequent drift claims. Pharmaceutical regulators already require post-approval stability studies that document product characteristics at specific timepoints; the principle transfers directly. Without a preserved reference baseline, all drift monitoring is self-reported and unverifiable.

Drift incident classification taxonomy. We need a standardized taxonomy of drift-related incidents, analogous to the cybersecurity incident classification frameworks that have enabled consistent reporting and learning across organizations and jurisdictions. At minimum, this taxonomy should distinguish: (Level 1) detected distributional change within pre-specified thresholds, documented but no intervention required; (Level 2) detected change exceeding alert thresholds, requiring internal review and documented response; (Level 3) confirmed performance degradation affecting system outputs in ways material to decisions, requiring deployer notification and regulatory disclosure; (Level 4) confirmed performance degradation associated with adverse outcomes, requiring immediate intervention, incident reporting, and root-cause investigation. The specific thresholds within each level are necessarily domain-specific, but the taxonomy itself should be standardized across deployers and jurisdictions.

Accountability chain for drift response. Every high-risk AI deployment must have a defined authorization chain for drift response decisions: who is notified at each alert level, who conducts the review, who is authorized to decide that intervention is required, and who can authorize suspension of the system's operation pending recalibration or retraining. This chain should be documented before deployment and available to the deployer's regulators. It should specify not only who has authority but who has accountability — meaning who bears documented responsibility if an alert is generated and no response occurs. The absence of defined accountability chains is precisely why, in the clinical scenario that opened this paper, a drift event that produced systematic harm for a patient subpopulation produced no intervention: nobody was specifically charged with making the call.

Public summary reporting. Periodic public reports on drift monitoring results for high-risk AI systems should be required — not at a level of proprietary technical detail, but sufficient to

enable external accountability. The report should state: whether significant drift was detected in the reporting period, at what level (using the incident taxonomy above), what response was taken, and whether the system's performance on defined subgroups remained within acceptable bounds. The analogy is bank stress test reporting: the methodology is disclosed, the high-level results are public, the proprietary details of the institution's internal models are not. This reporting enables regulators, researchers, and affected communities to identify patterns across deployers and sectors that no single organization's internal monitoring would surface.

7. Recommendations

We close the analytical section of this paper with five specific recommendations, each directed at named actors with specific actions. These are not aspirational principles. They are concrete steps that can be taken under existing institutional mandates, without waiting for broader legislative reform.

Recommendation 1 — For the European Commission: Operationalize Article 72 with Drift-Specific Requirements

The implementing act establishing the template for post-market monitoring plans under Article 72 of the EU AI Act must include mandatory sections specifying: (a) the drift types relevant to the system; (b) the monitoring methodology and signals for each drift type; (c) the alert thresholds; and (d) the response protocols. A monitoring plan template that permits developers to describe their monitoring in generic terms — "we monitor system performance" — is not a monitoring standard. The Commission should draw on the technical literature on dataset shift detection and the FDA PCCP framework as inputs to drafting standards that require substantive specification.

Recommendation 2 — For NIST: Develop Sector-Specific Drift Monitoring Profiles

The NIST AI RMF should be extended with sector-specific profiles that translate its generic MEASURE function provisions into concrete drift monitoring requirements for high-risk deployment contexts. Profiles for healthcare, criminal justice, and financial services should specify: which drift types are of primary concern in that sector; what monitoring signals are appropriate; what constitutes a meaningful shift worthy of investigation; and what documentation standards apply. These profiles should be developed in collaboration with domain regulators (FDA, CFPB, DOJ) and should be positioned as the minimum adequate implementation of the MEASURE function in their respective sectors, even under the voluntary framework.

Recommendation 3 — For the FDA: Extend the PCCP Framework to Cover Unintended Performance Change

The FDA's Predetermined Change Control Plan framework should be extended to require a companion "Performance Monitoring Plan" (PMP) that specifies, analogously to the PCCP's Description of Modifications, the specific monitoring signals, thresholds, and response protocols for unintended performance changes — including drift under each of the three statistical drift types relevant to classifier-based medical AI. The PMP should be reviewed as part of the

marketing submission and updated with each major software version. This extension does not require new statutory authority — it is a natural extension of the PCCP framework's logic to the unintended-change domain.

Recommendation 4 — For Standards Bodies (ISO/IEC JTC 1/SC 42): Develop a Technical Standard for AI Drift Monitoring

ISO/IEC should develop a dedicated technical standard for AI behavioral drift monitoring — distinct from the management system level requirements of ISO 42001 — that specifies: (a) standardized methods for establishing and preserving reference baselines at deployment; (b) statistical methods appropriate for detecting each drift type; (c) a minimum incident classification taxonomy; and (d) documentation requirements for drift monitoring results. Such a standard would provide the technical substrate that regulatory frameworks like the EU AI Act and NIST profiles currently lack, enabling auditors to evaluate whether monitoring is genuinely adequate rather than formally present.

Recommendation 5 — For Deployers in High-Risk Sectors: Implement Pre-Deployment Drift Specification Now, Without Waiting for Regulation

Hospitals, financial institutions, criminal justice agencies, and other deployers of high-risk AI systems should not wait for regulatory requirements to implement the minimum governance architecture described in Section 6. Pre-deployment drift specification and reference baseline preservation can be implemented today, using existing MLOps tooling and standard statistical methods. Organizations that do this now will be ahead of the regulatory curve; those that do not will find themselves retroactively applying governance to systems that have been operating without it — a more difficult and more legally exposed position. We recommend that deployers adopt the drift incident classification taxonomy proposed in Section 6 as an internal standard, regardless of whether external frameworks currently require it.

8. Conclusion

Drift is the point where AI systems and the world diverge. Every model that is trained is, in a sense, a snapshot of a world-state — a statistical summary of the relationships that existed in the data at the time of training. The world keeps moving after the snapshot is taken. Population demographics shift. Economic conditions evolve. Clinical practices change. The adversarial landscape expands. And the model, which cannot update itself, keeps applying its frozen understanding of a receding past to an ongoing present.

All of our governance frameworks are built on the implicit assumption that the world the model was trained on is the world the model will encounter. That assumption is most dramatically wrong precisely where the stakes are highest. In healthcare, where models trained on historical patient populations encounter the living, changing demographic reality of communities. In criminal justice, where recidivism prediction models encode a snapshot of the relationship between observable features and reoffending that reflects a specific historical configuration of policing, poverty, and social conditions. In financial services, where credit risk models trained on pre-crisis data are deployed into post-crisis economic landscapes that have structurally

reorganized the relationships those models depended on. In critical infrastructure, where predictive maintenance models trained on nominal operating conditions may encounter conditions that have shifted in ways that make nominal-trained predictions systematically wrong.

We are not arguing that these frameworks were designed in bad faith. We are arguing that they were designed without fully confronting the implication of deploying statistical learning systems in a changing world. The evaluation paradigm that underlies almost all AI governance — evaluate the system, establish that it meets the standard, certify it for deployment — is a paradigm inherited from hardware and software certification, where the thing being certified doesn't change its behavior based on what it encounters. AI systems are different in kind, not just degree. A statistical learning system is not a bolt with a rated load capacity. Its effective behavior is a function of the world it encounters, and the world changes.

The governance mechanisms we described in Section 6 are not technically exotic. Reference baseline preservation is a straightforward data management practice. Pre-deployment drift specification is a form of pre-registration that the scientific community has used for decades. Incident classification taxonomies are standard in cybersecurity and aviation. What they require is institutional will — the willingness of regulators to specify requirements that create real obligations, of developers to implement monitoring that generates real accountability, and of deployers to treat drift governance as a core operational responsibility rather than a compliance formality.

The difference between governance frameworks that produce accountability and those that produce paperwork is whether they specify, in advance, what happens when something goes wrong — who discovers it, who decides, who is responsible, and how it is documented. On behavioral drift in AI systems, we have almost entirely produced paperwork. The patients triaged by a drifted risk score, the loan applicants scored by a miscalibrated credit model, the defendants assessed by a concept-drifted recidivism predictor — they deserve better. So does the governance enterprise itself, which cannot credibly claim to manage AI risk while ignoring the most structurally inevitable source of post-deployment performance change that statistical learning systems face.

We publish this paper with the hope that it is useful to the practitioners, regulators, and researchers who are working in good faith to close these gaps. The problem is tractable. The tools exist. The institutional will to use them is what is missing.

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